FIELD GUIDE · AGENTIC ENGINEERING

The workspace is the product.

Models are rented. Harnesses are swappable. What compounds is the place they work in: its instructions, memory, procedures, structure, and the colleagues beside them.

SEVEN ACTS · ABOUT FORTY-FIVE MINUTES · PRESS → TO BEGIN, G FOR THE GUIDE



THE THESIS

Give the agent a filesystem and a shell.

A model guesses when it can't look. A workspace lets it look, run, and write down what it found.

Chat only | Filesystem + shell

Explicit. Every input is a file you can open, diff and cite.

Open. Bash runs whatever the harness allows: search, tests, scripts.

Durable. What it learns is written back, so the next run starts ahead.

Structure and philosophy decide how it works there. That is harness engineering.



THE OPERATING MODEL

Seven questions. One system.

Each layer answers one question an agent asks before it can do useful work.

01

Workspaces

Where does it live? Primer, policy, record and sandbox.

02

Context

What reaches the window, and how hooks make recall a reflex.

03

Second brain

How the graph is structured, read and grown.

04

Skills

How we do this kind of work, loaded on demand.

05

Structure

Which bytes it may touch, and where they go.

06

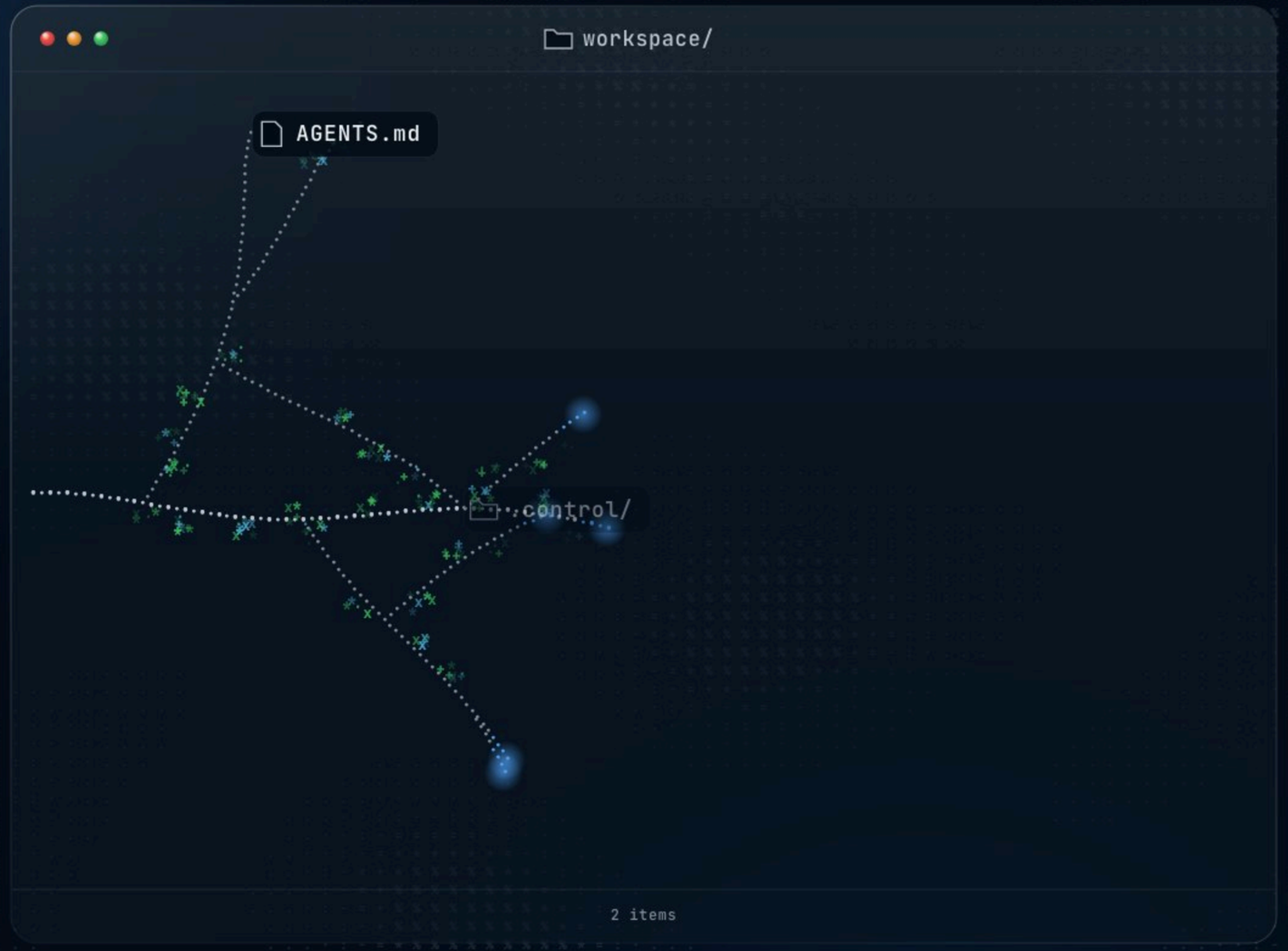
Fleets

Who else is here, and who checks whom.

07

Leadership

What all of this asks of people.



ACT I

Workspaces

Where does the agent live?

— I · WORKSPACES

A folder is not a workspace.

The difference is what survives the session.

Agent workspace | Agentic workspace

AGENT WORKSPACE

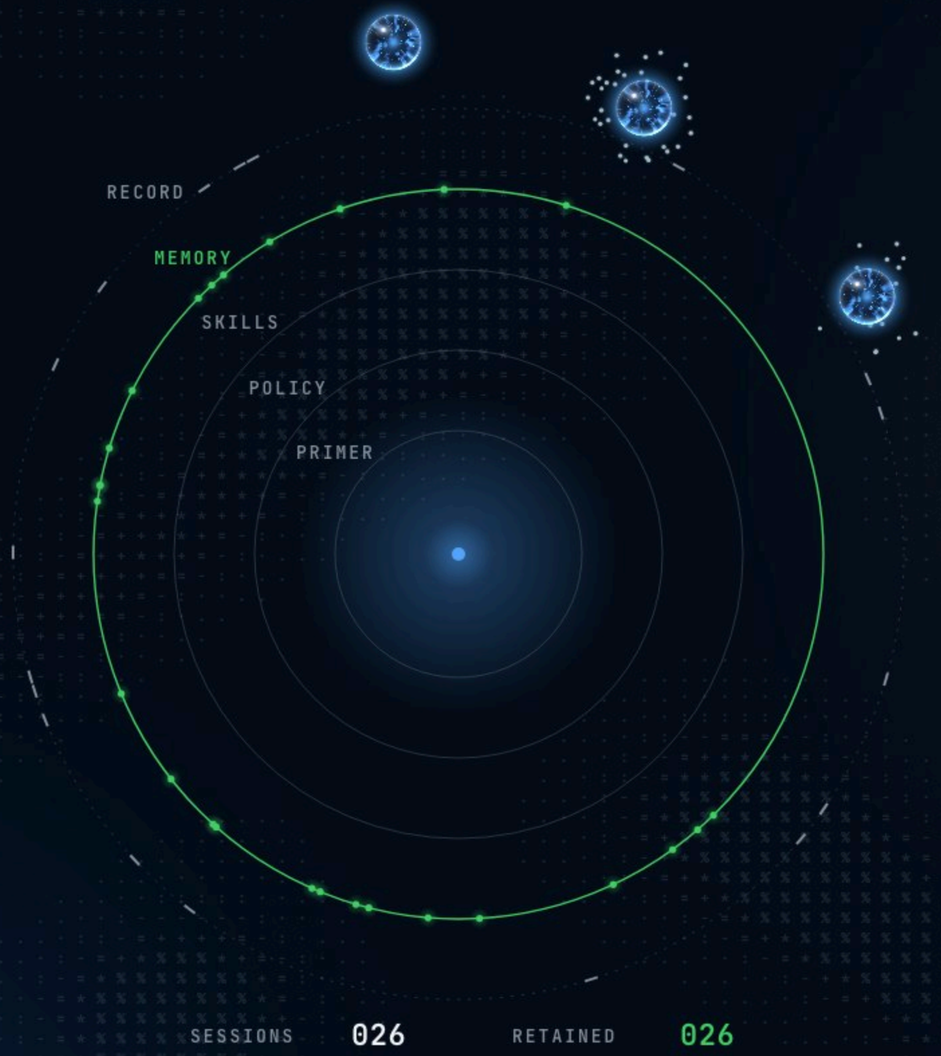
A working directory and a toolset for one agent on one task.

Useful. But context evaporates when the session ends, and the next agent starts cold.

AGENTIC WORKSPACE

A governed place that many agents work through.

Instructions, policy, memory, skills and a record outlive every session, so each run leaves the place a little smarter.



— I · WORKSPACES

Six layers an agent reads before it writes.

LAYER 01

Primer

AGENTS.md ≡ CLAUDE.md

Philosophy, routes and hard boundaries. One file, byte-identical copies for every harness.

RULE Route, don't dump. Point to protocols instead of inlining them.

READ ORDER ↓

01	Primer	AGENTS.md ≡ CLAUDE.md
02	Policy	.control/policy.yaml
03	Skills	skills/<category>/<name>/
04	Knowledge	research/entities/ · knowledge-index.md
05	Record	docs/conversations/ · .control/initiatives/
06	Work	repos · worktrees · sandboxes

WRITE PATH ↑

— I · WORKSPACES

Sandbox, harness, wire.

Three layers, three different answers. Buy the box, swap the brain, and don't bet on the wire yet.

Claude Code | Codex | Gemini CLI | Your own

UNSOLVED

Wire

phone · desktop · chat

SWAP

Harness

the agent loop

BUY

Sandbox

microVM · persistent volume



MCP SERVER + CLI — the integration point never moves when the harness does

— I · WORKSPACES

Medallion layers live in the filesystem.

Each workspace keeps its own bronze, silver and gold as plain folders. A shared ledger records what was decided.

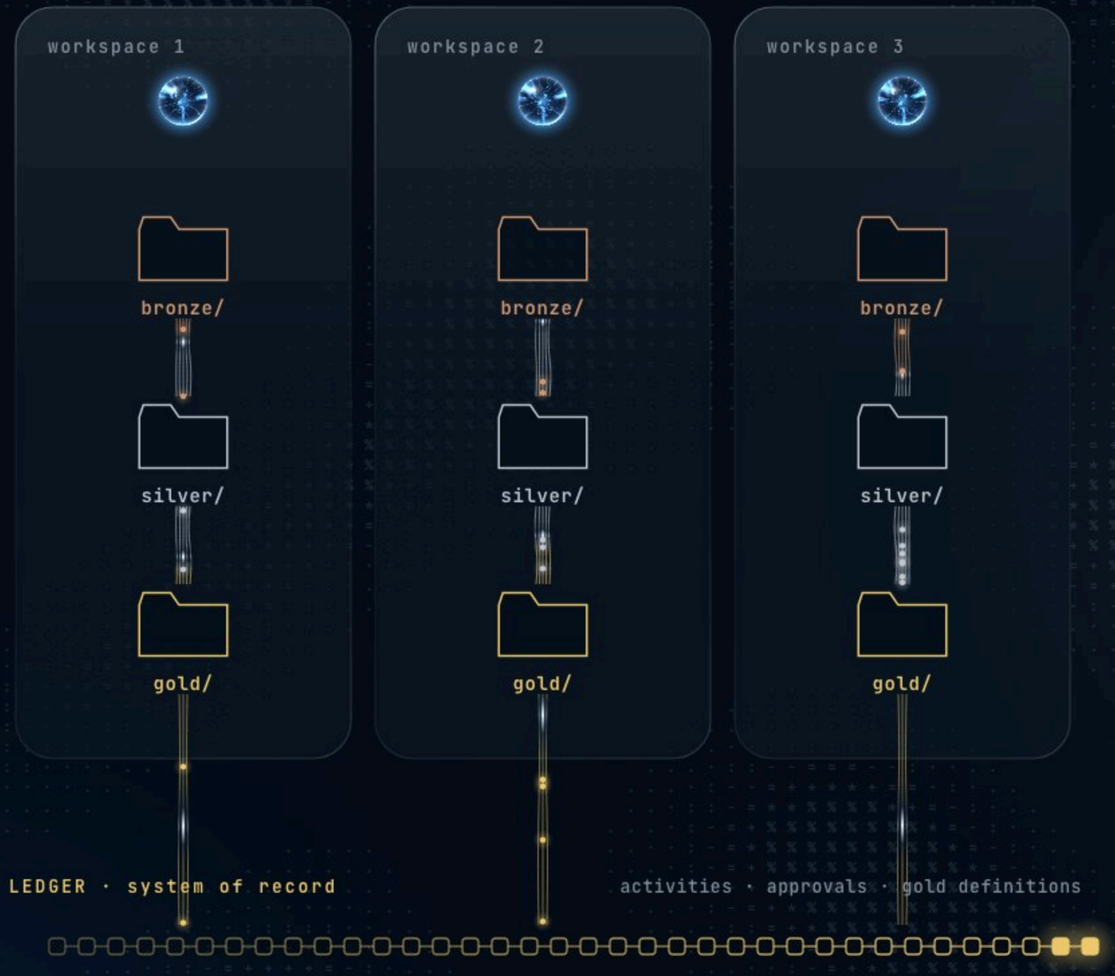
One workspace | Many workspaces

bronze/ Inputs as received. Append-only, hashed, never edited.

silver/ Typed and validated. Every value cites its bronze source.

gold/ Decisions and definitions the organization trusts.

Files are the agent's working memory. The ledger is the organization's.



— I · WORKSPACES · WORKED EXAMPLE

A review system, built on the pattern.

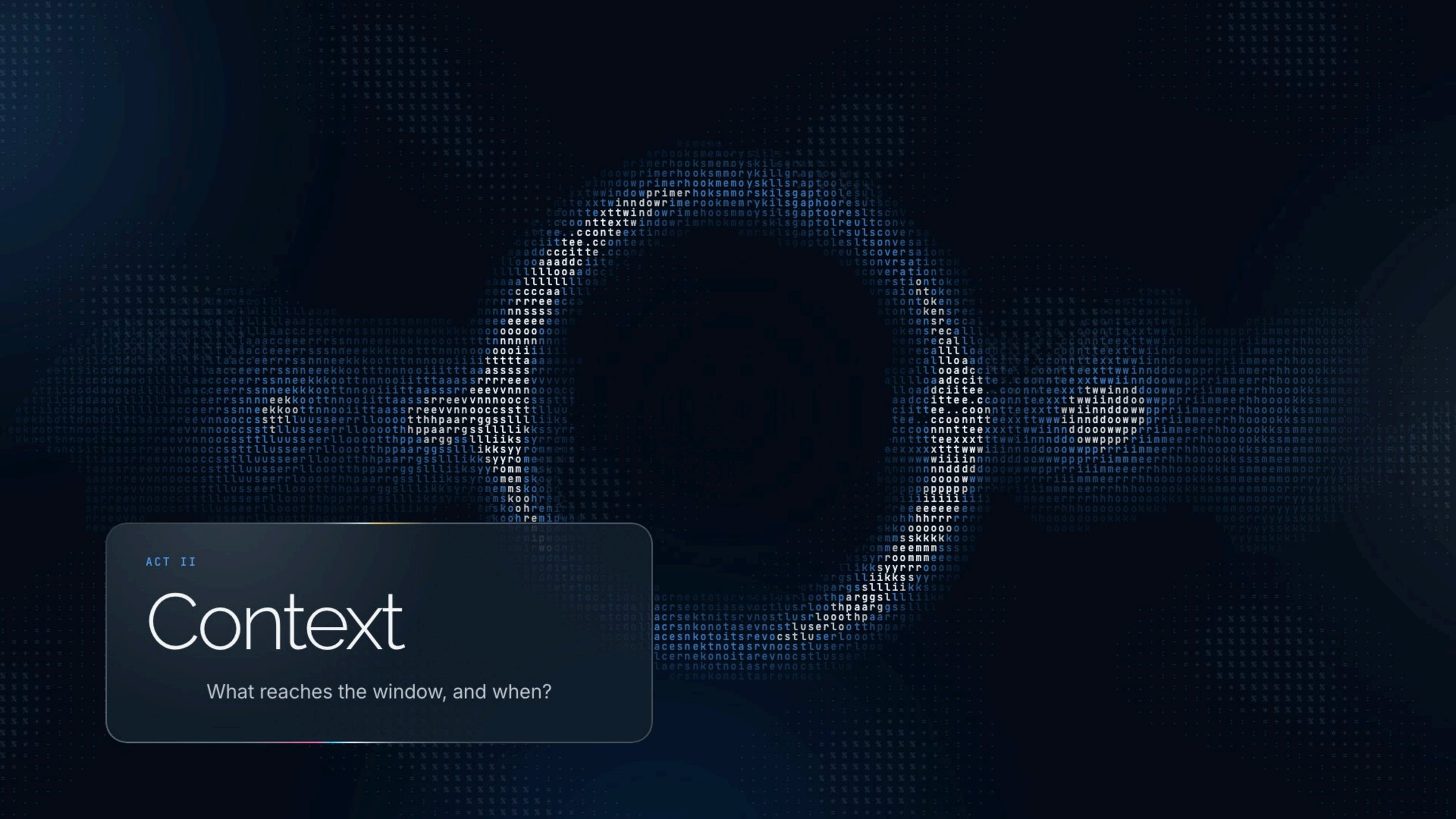
Sandbox | Readers | Judges | Ledger

● BEAT 04 · LEDGER

The ledger is the system of record

Activities, approvals and gold definitions go to one shared Postgres ledger outside every sandbox. Files are the working memory; the ledger is what the organization trusts.




ACT II
Context
What reaches the window, and when?

— II · CONTEXT

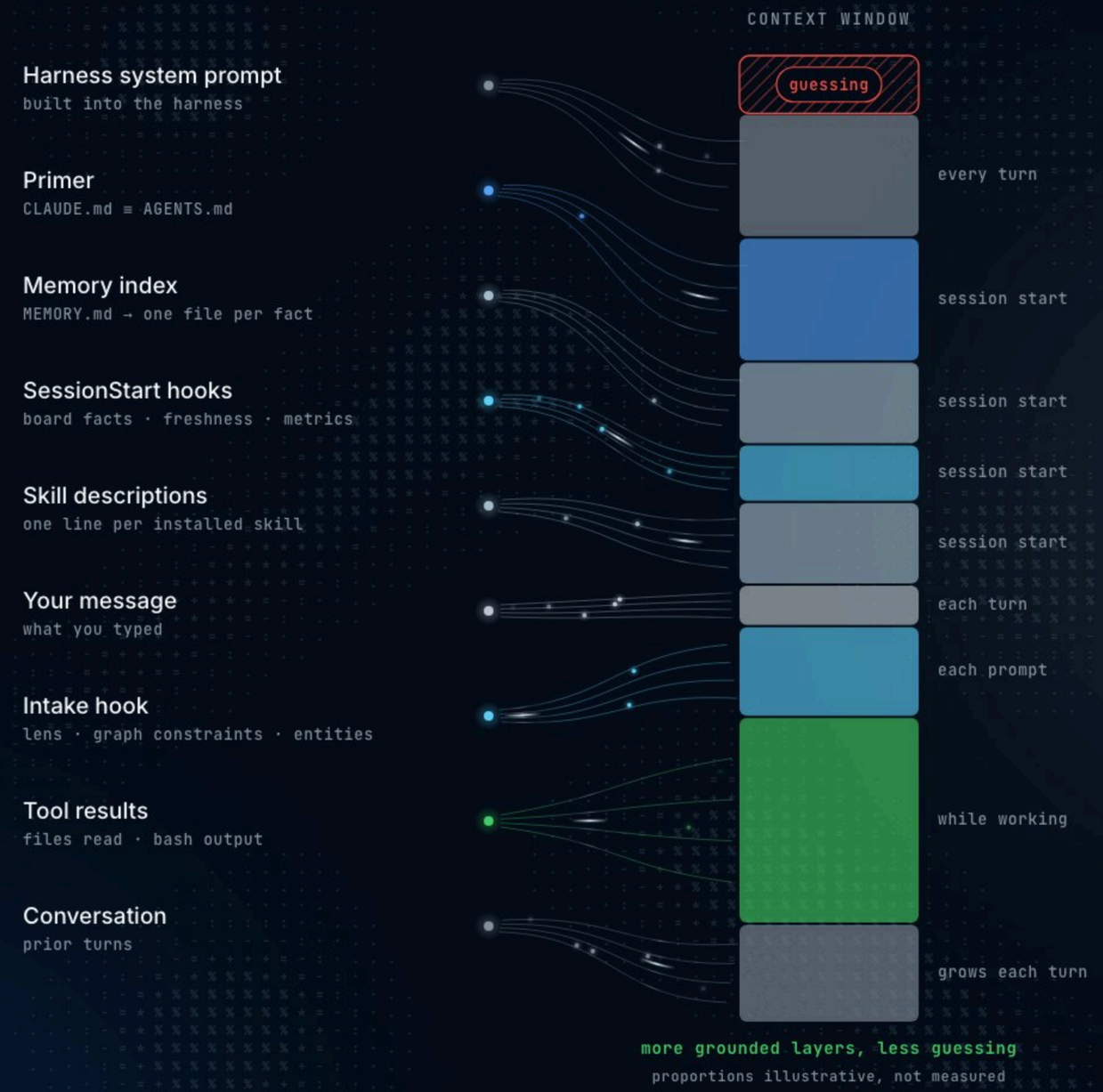
How an agent's context is built.

Everything the model knows this turn is what reached the window. The workspace decides what reaches it.

Bare session | Workspace session

Hover any layer of the window. Each one has a source on disk, a moment it loads, and a budget.

Less guessing is the product of more grounded layers, not of a bigger model.



II · CONTEXT

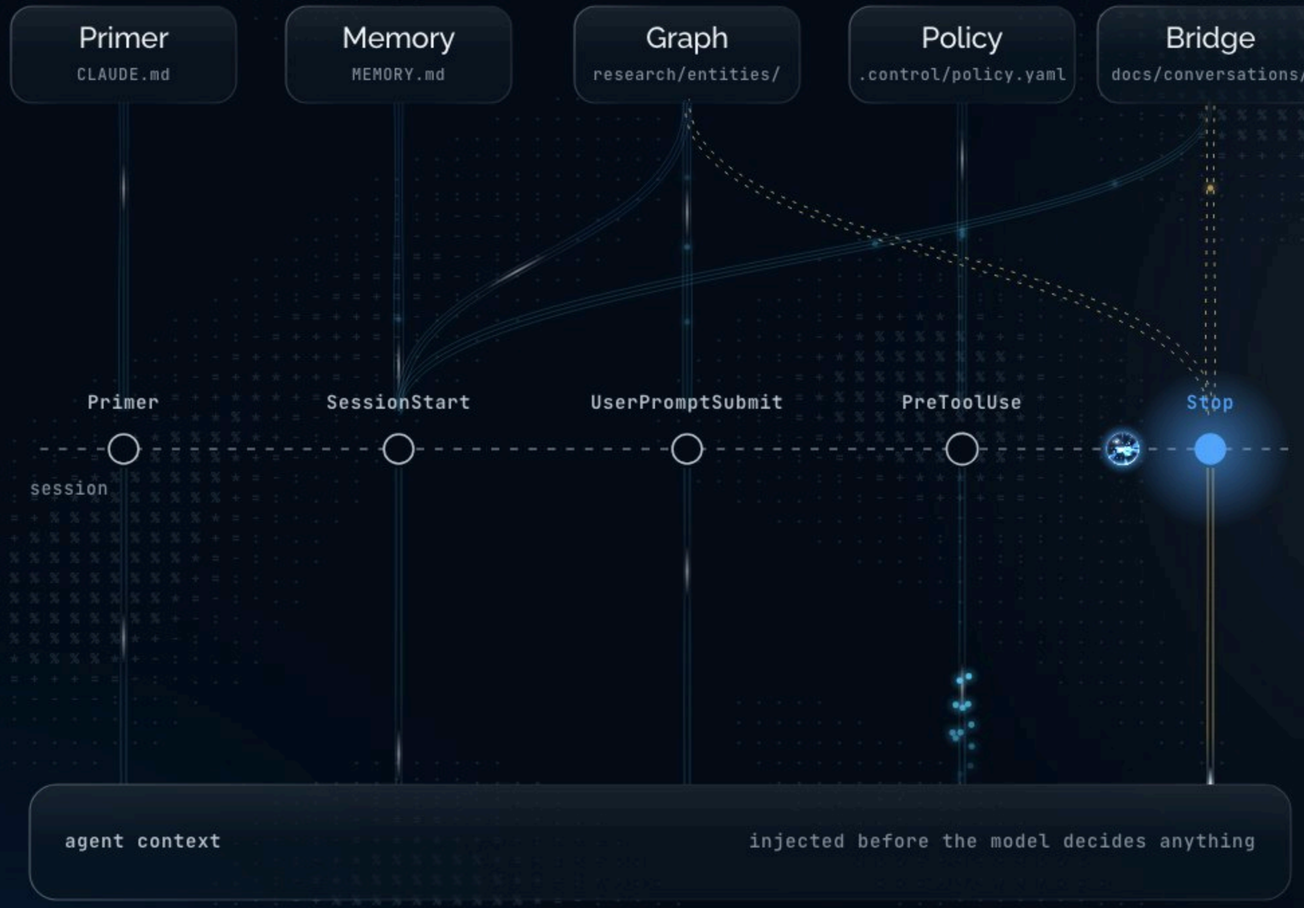
Hooks turn recall into a reflex.

HOOK 05 OF 05 · BRIDGE · PROVENANCE

Stop

When the session stops, the bridge writes it to docs/conversations and provenance is stamped, so the next session can search what this one did.

`docs/conversations/session-2026-10-26-….md`



PRIMER
Asks for it

SKILL
Knows how

HOOK
Runs every time

— II · CONTEXT

Files first. Vectors when words diverge.

All | Local | Semantic | Live

Exact names, paths, claims you can grep	RG + INDEX
What this relates to, one hop out	GRAPH LINKS
Same idea, different words; corpora past the window	EMBEDDINGS
Fresh or external state	MCP · APIS



At 1,430 entities the compiled catalog is about 260K tokens: it fits a 1M-token window, so the catalog routes and the model reads. Add vectors when vocabulary diverges or the catalog outgrows the window.

ACT III

Second brain

How is knowledge read, and how does it grow?

@ the reader

— III · SECOND BRAIN

Most second brains are write-only.

Today | Wired at intake

● THE FIX IS ROUTING, NOT VOLUME

Put recall on the startup path.

The primer routes to it. An intake hook surfaces matching entities. `/kg load` runs before planning. And when reads lag, prune production. Don't add more writers.

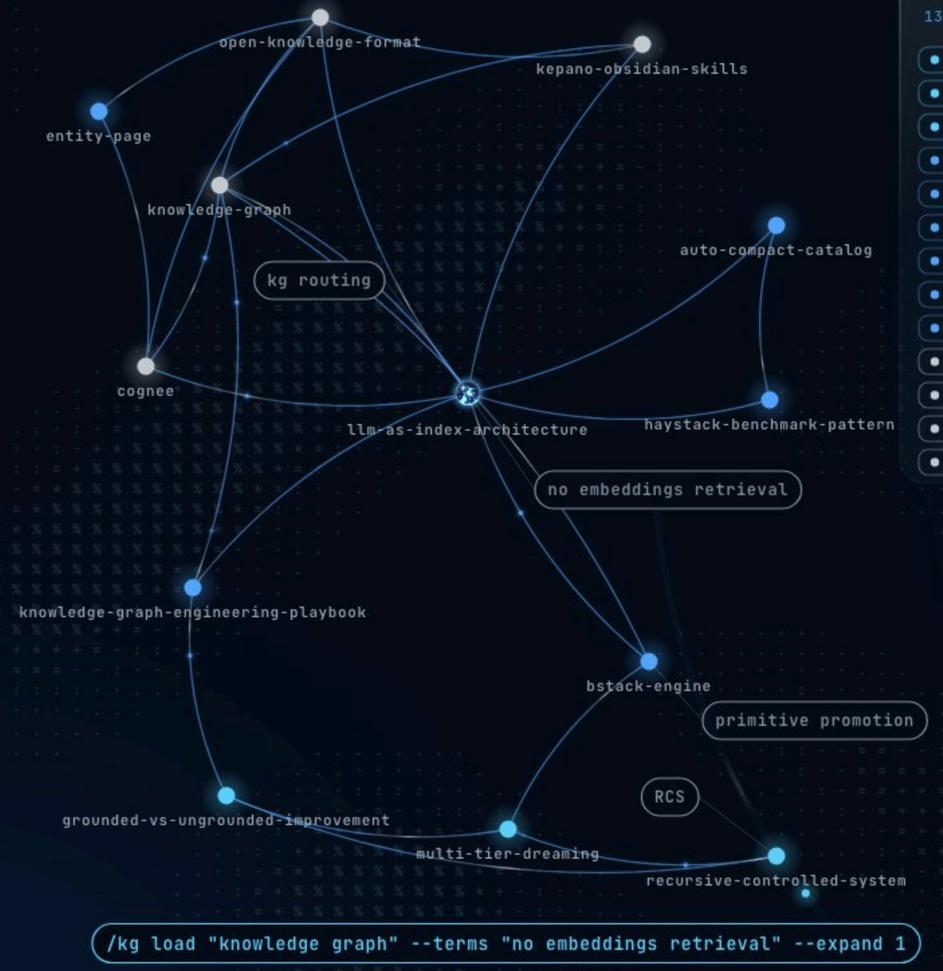


— III · SECOND BRAIN

Front matter is the API.

Files | Index | Edges | Synonyms

```
research/entities/pattern/
llm-as-index-architecture.md
---
type: pattern   status: entity
core_claim: "At sub-10k-entity scale the
  substrate fits in a 1M-context
  model, so the LLM IS the query…"
tags: [architecture-decision, knowledge-graph, …]
aliases: [kg routing, no embeddings retrieval]
related: [[knowledge-graph]], [[bstack-engine]] …
sources: [2026-05-21-…-synthesis, BRO-1223]
---
```



III · SECOND BRAIN

An entity is a claim with receipts.

```
---
# research/entities/concept/agent-harness-three-layer-split.md
type: concept   status: entity
core_claim: "Agent harnesses split into sandbox,
  orchestrator and UI wire: the sandbox
  is buyable, the wire has no stable
  remote transport."
sources:
  - url: agentclientprotocol.com/…/transports
    note: "Read verbatim 2026-09-26 …"
scoring: {novelty: 2, specificity: 3, relevance: 3}
related: [[buzz-acp-harness]], [[display-server-for-agents]] …
---
```

One falsifiable sentence

Read verbatim, dated

Scored before promotion

Typed links become edges



1,430 entities 6,076 edges .md canonical · index of 2026-10-25

— III · SECOND BRAIN

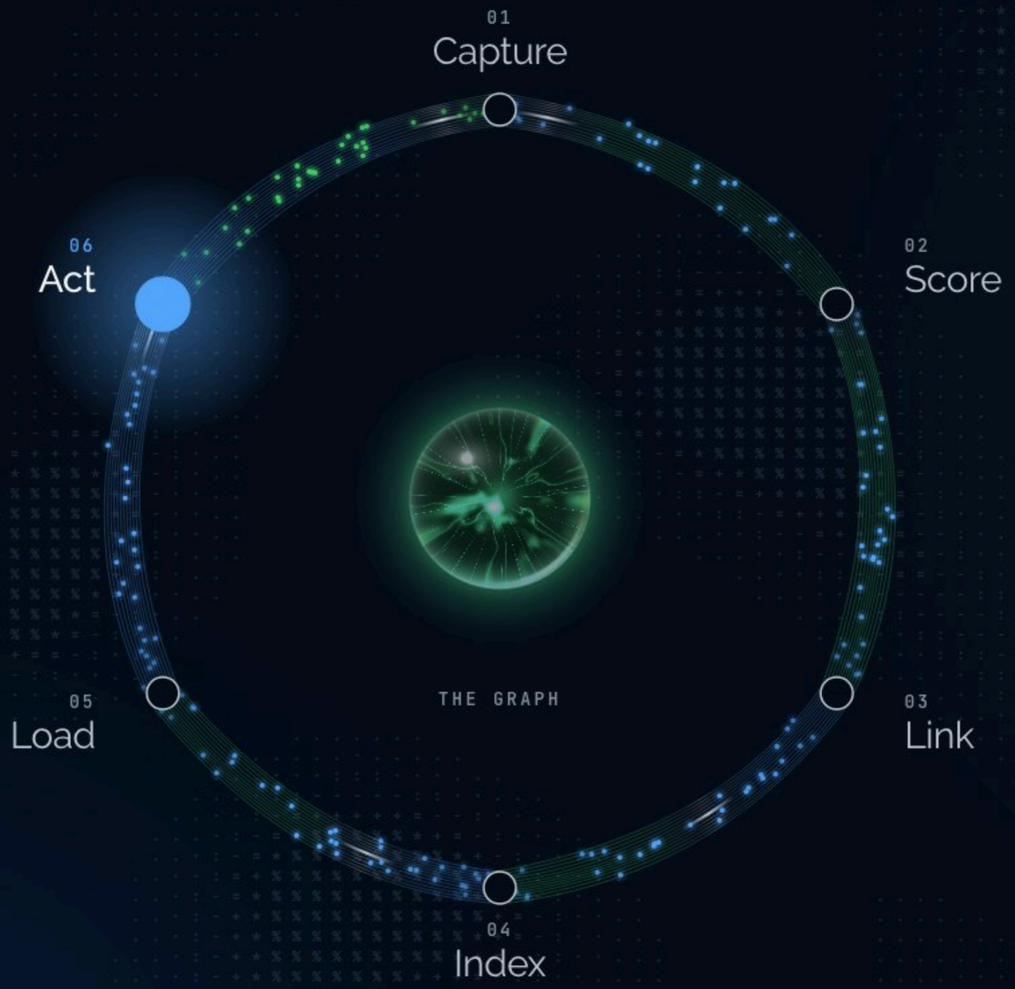
The loop that makes knowledge readable.

● STATION 06 OF 06

Act

`cite → decide → capture`

Outputs cite entities. New findings re-enter the loop.



Memory holds facts about the owner. The graph holds claims about the world. The ledger holds what is in flight.

III · SECOND BRAIN

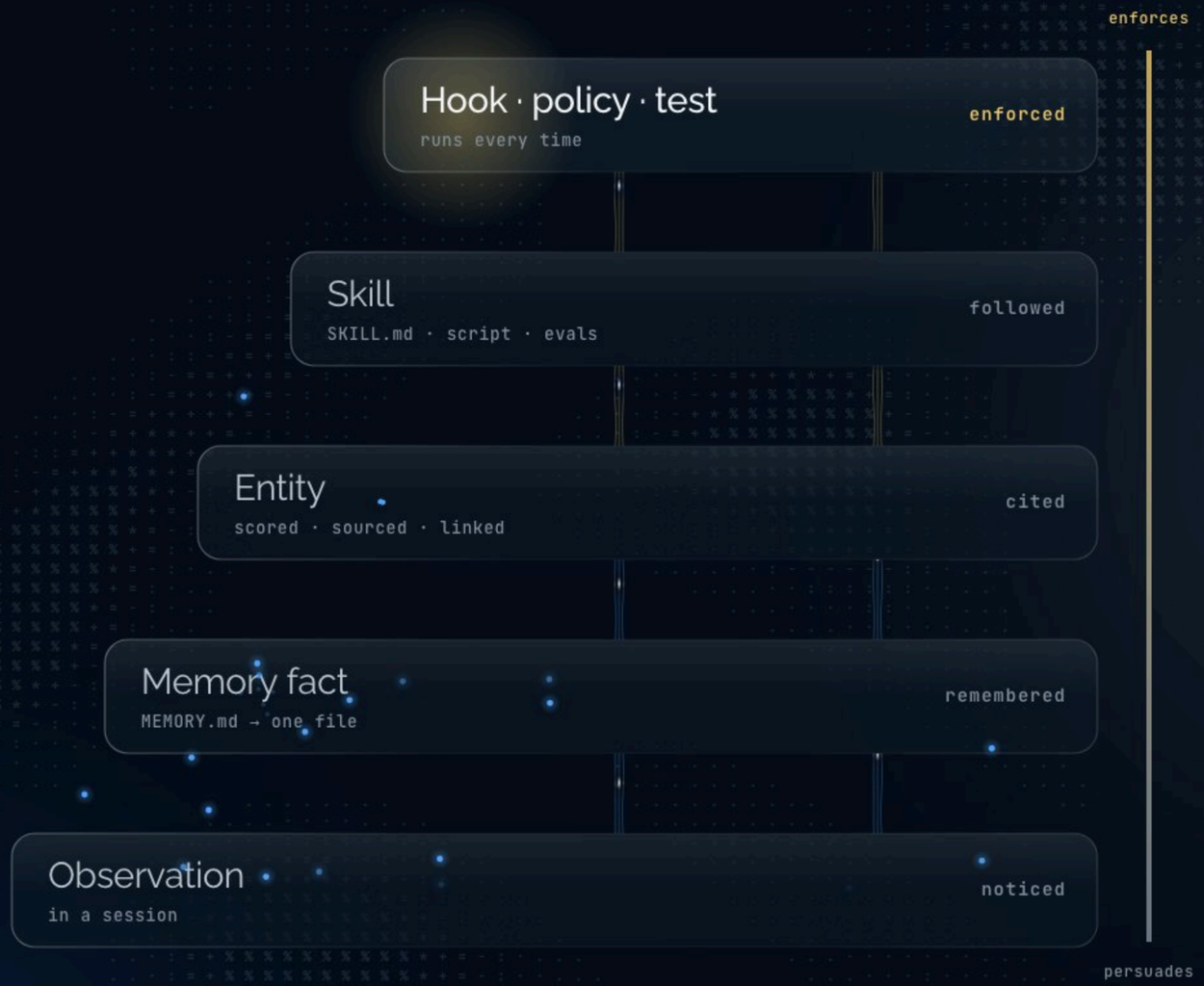
Insight hardens into mechanism.

LEVEL 05 OF 05 · ENFORCED

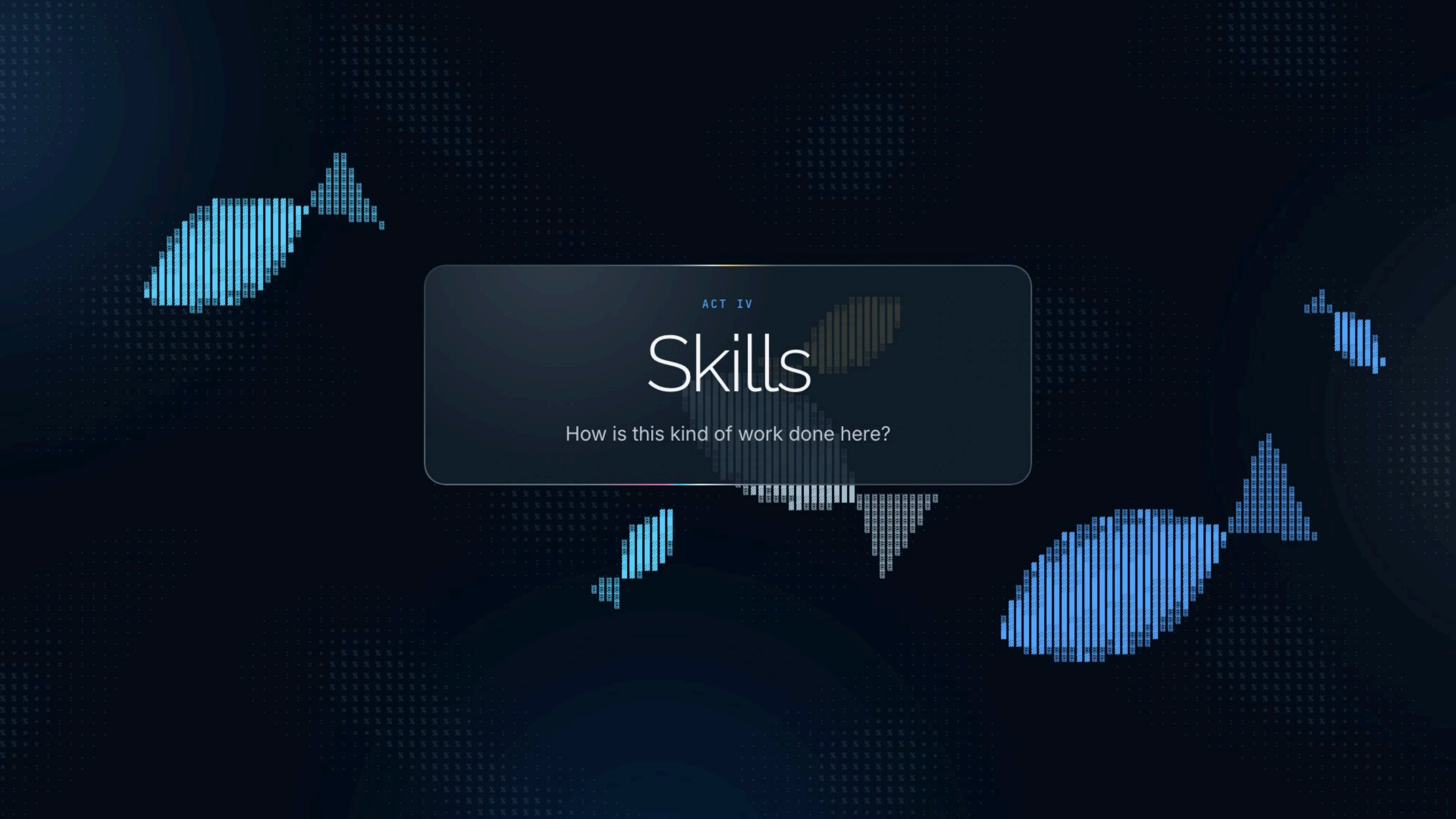
Hook · policy · test

A rule that must never be broken stops depending on anyone remembering it. It executes.

Prose persuades. Executables enforce.



Promote on the third instance, with a concrete mechanism, an invariant and a failure mode. Name what the new rule retires.


ACT IV
Skills
How is this kind of work done here?

— IV · SKILLS

A skill is a procedure, not a prompt.

```
skills/knowledge/kg/
├─ SKILL.md       description is the router
├─ scripts/       deterministic core
└─ evals/         routing + behavior tests

---
name: kg
description: Load graph context for a topic…
  USE WHEN … NOT FOR …
---
```



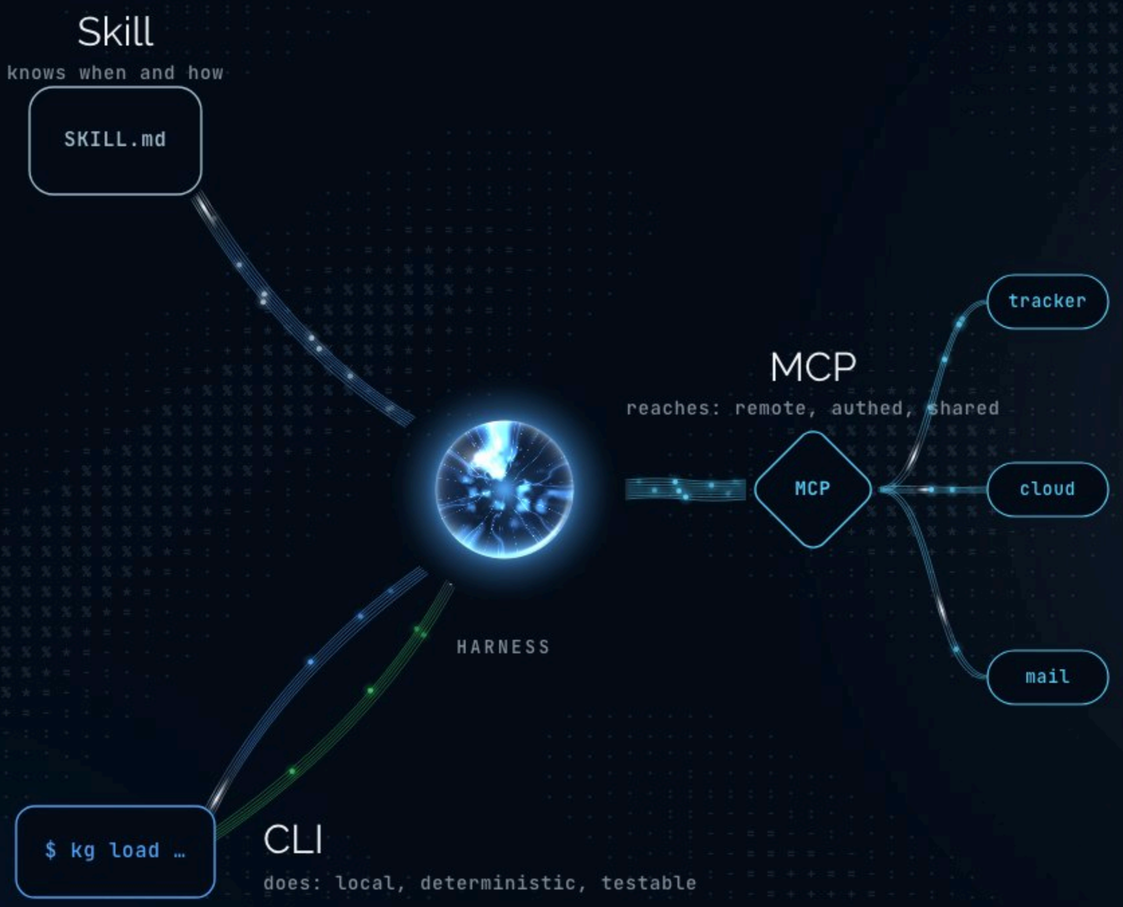
— IV · SKILLS

Skill, CLI, MCP.

One knows. One does. One reaches.

All | Skill | CLI | MCP

Repeatable judgment and steps	SKILL
Deterministic, local, testable work	CLI
Remote system, auth, shared across harnesses	MCP SERVER
A one-off	JUST ASK



A skill that drives a small CLI beats a wide MCP surface: fewer tools in context, and a core you can test.

— IV · SKILLS

Author once. Fan out. Ration hard.

Fan out | Ration

115 AUTHORED

23 CATEGORIES

31 HARNESS FOLDERS

- One canonical source, in its own repo.
- Install once, symlink into every harness. Never copy.
- New skills ship evals: it fires when it should and stays quiet when it shouldn't.
- Retire before you add. The roster is a budget.

457 INSTALLED

26 KEPT A DESCRIPTION



the rest were bare names: they could not trigger



ACT V

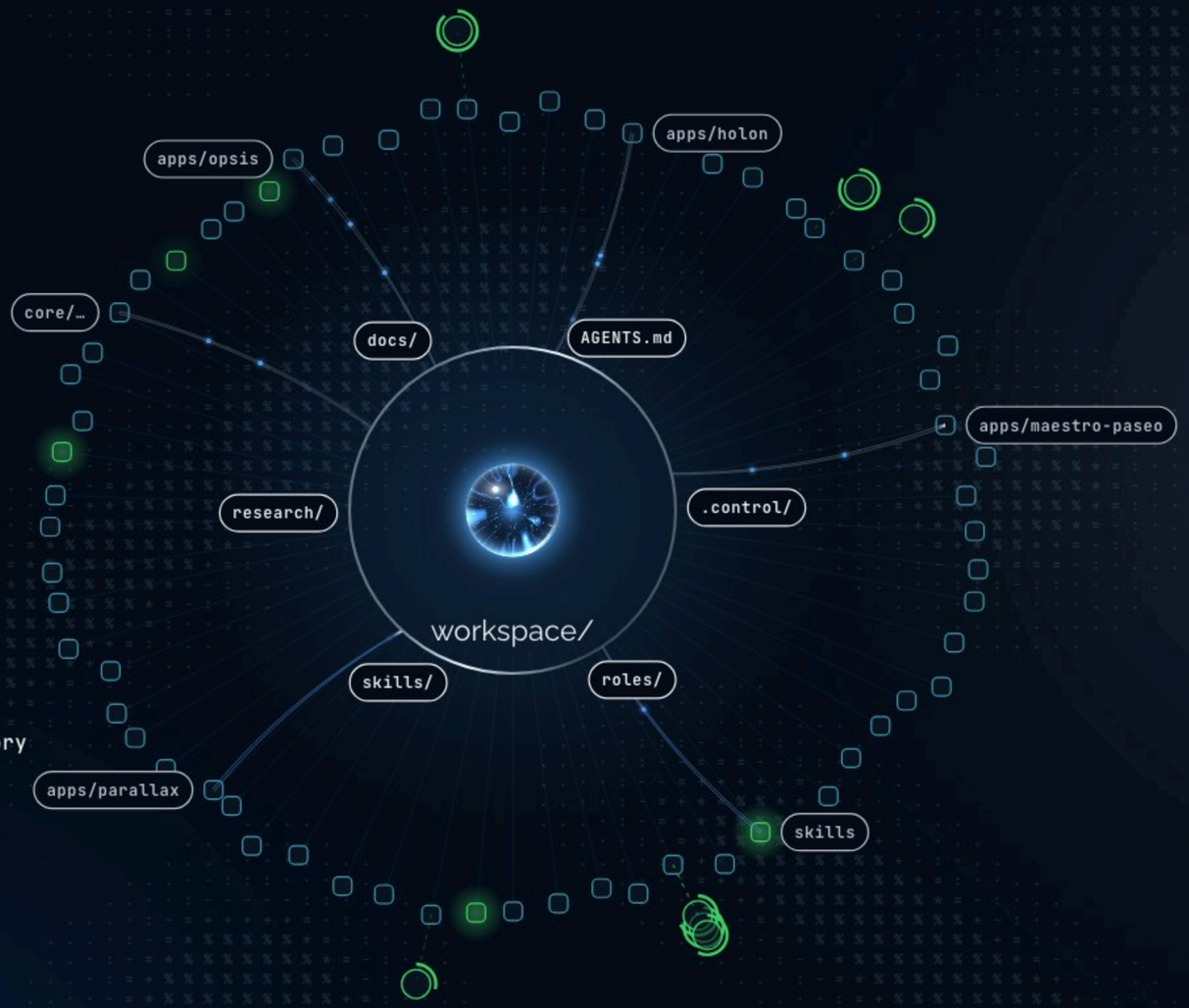
Structure

Which bytes may the agent touch?

— V · STRUCTURE

One place, many repos.

workspace/	governance repo
├─ AGENTS.md ≡ CLAUDE.md	one primer, every harness
├─ .control/	policy · initiatives · asks
├─ roles/	lenses: context per kind of work
├─ skills/	canonical skills, own repo
├─ research/entities/	the graph, as markdown
├─ docs/	sessions · plans · handoffs · ADRs
├─ apps/ core/ work/ …	product repos, own CI
└─ .worktrees/	one per task, disposable



- **SHARED** — one copy of the operating context, reviewed like code
- **SOVEREIGN** — 66 product repos under apps/, core/ and work/, each with its own history
- **DISPOSABLE** — worktrees created 7 · merged 0 · swept 0

V · STRUCTURE

One writer per tree.

Parallel agents never share a checkout. They share main, through review.



01
Branch is the first action

02
Reviewers read. They never write

03
A live thread's worktree is the thread

04
Unfinished work becomes a draft PR

05
Janitor after merge, dry-run first


ACT VI
Fleets
Who else is here, and who checks whom?

VI · FLEETS

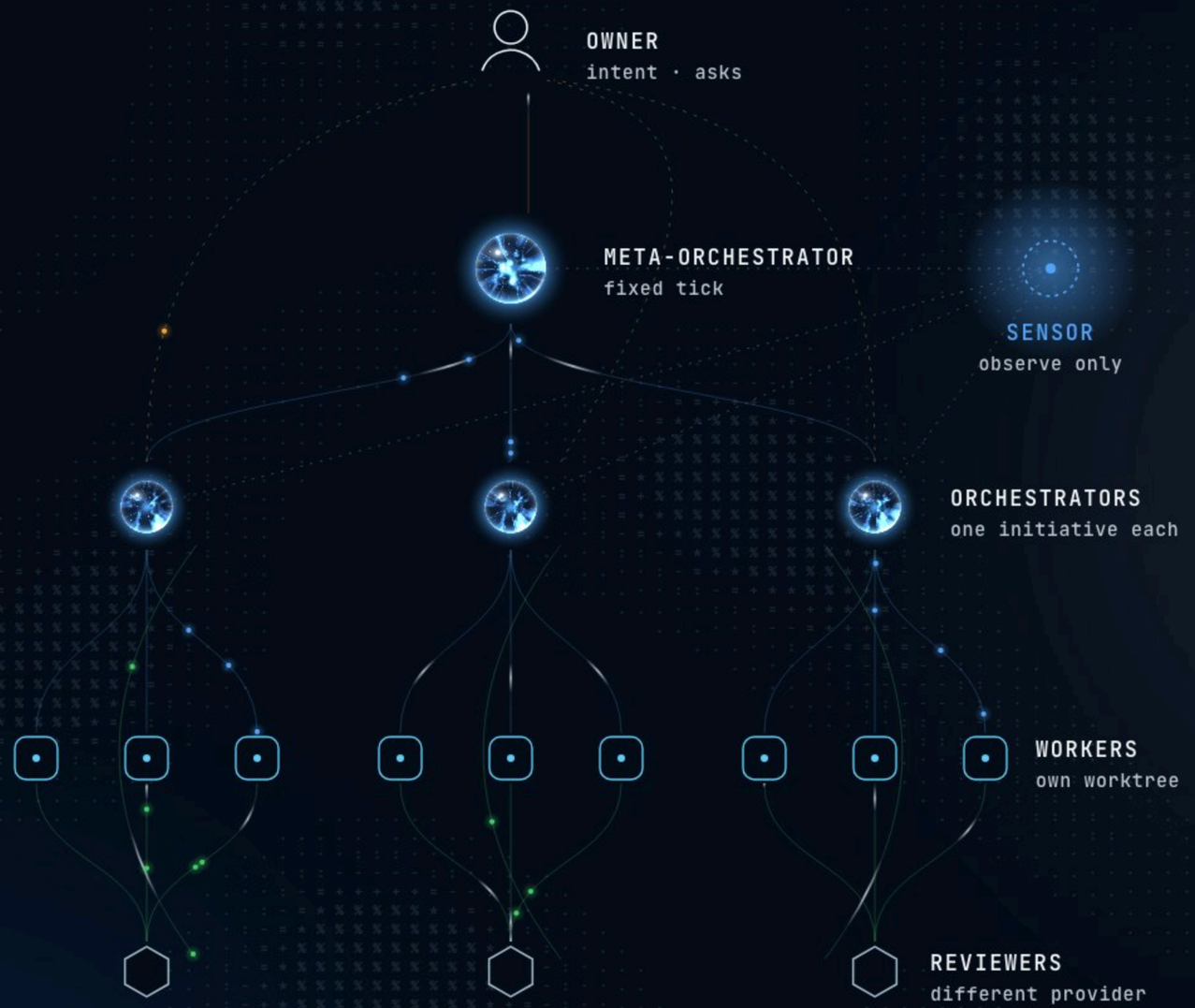
Roles, not agents.

ROLE 06 OF 06

Sensor

Observes and files: capacity, drift, stalls.

NEVER Acting on what it measures.



VI · FLEETS

Spawn, reuse, or hand off.

Independent of what I know?	Spawn a child task in its own worktree.
Needs my context?	Continue here, or fork the thread.
Outlives this context window?	Write a persistent brief with stop conditions.
Only the owner can decide?	File an ask: context, options, a recommendation and a safe default.



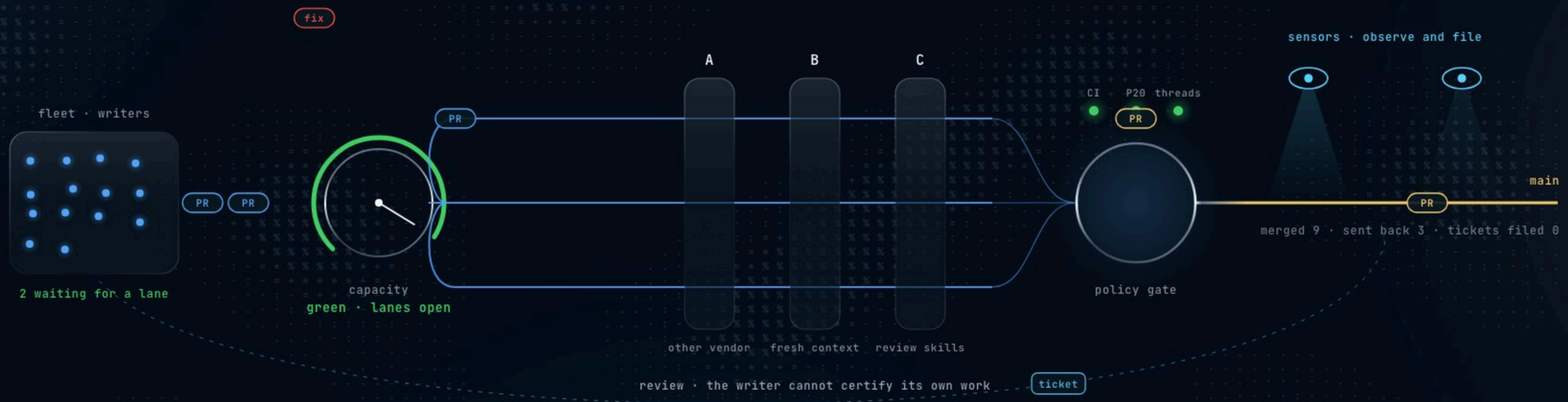
Long sessions carry most of the token cost. Roll over with a handoff.

Handoff	Handback	Brief	Ask	Receipt
narrative for the next agent	message for the human	loop state across contexts	a decision, with its context	evidence it actually works

VI · FLEETS

Governance that scales with the fleet.

Pace | Review | Policy | Sensors



PACE
From measured capacity, not hope

REVIEW
The writer cannot certify its own work

POLICY
Executable gates. Prose is only advice

SENSORS
Observe and file. Agents act


load · act · capture · score · link · index
ACT VII
Leadership
What does this ask of people?

VII · LEADERSHIP

The unit of work is now a loop.

When an attempt costs minutes, making gets abundant and deciding gets scarce.

Task | Loop

Every question is something you can start today.

Every repeated task is something you can automate or improve.

Every piece of work leaves the system better: a note, a skill, a check.

The question moves from “can I do this?” to “should this exist, and how will I know it's right?”



VII · LEADERSHIP

Everyone leads a fleet.

TERM 05 OF 05

Review

who judges the result before it ships

You do not certify your own work: a separate session or a different model does.



What each person brings to their fleet is taste and judgment: knowing what good looks like, and deciding what to accept.

VII · LEADERSHIP

Attention is the new ration.

Every thread pings you | **The gate calls you**

Reversible drafts, research, branches: act and note it.

Costly long compute, many iterations: only inside a budget.

Irreversible or external spend, publish, merge outside policy, contact: a yes first, every time.

The measure of an autonomous system isn't how much it does without you. It's how precisely it calls you.



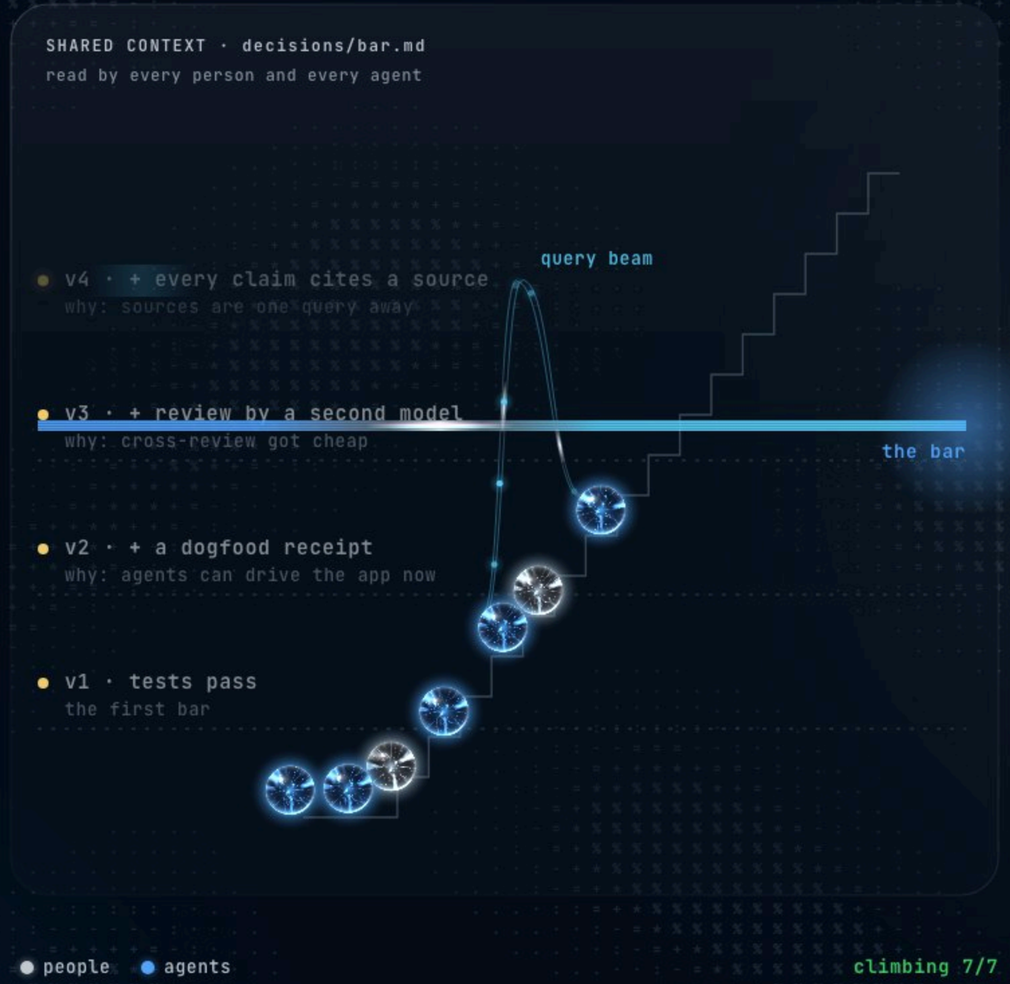
VII · LEADERSHIP

Goalposts teach waiting. Bars rise in the open.

Goalpost moves | **Bar rises**

People learn to wait when proposals get overridden, the leader decides anyway, the bar moves silently, or a failed attempt costs more than a passive miss.

Make everything queryable: decisions, standards and why they changed live in the same shared graph that people and agents read.



EPILOGUE

Better agents come from better harnesses.

LAYER 06 OF 06

Workspace

The place that holds all of it and keeps what every session learns. This is where the harness compounds.

MODEL — the same weights in every layer

OUTCOME — illustrative, not measured

evals · 78 passed · 13 failed

workspace · 009 deposits kept



EPILOGUE

Twenty primitives. One loop.



The workspace is the product.

01

Give agents a place before you give them more autonomy.

02

Make the graph readable at intake, or stop writing to it.

03

Scale with roles, isolation and review. Not with more prompts.